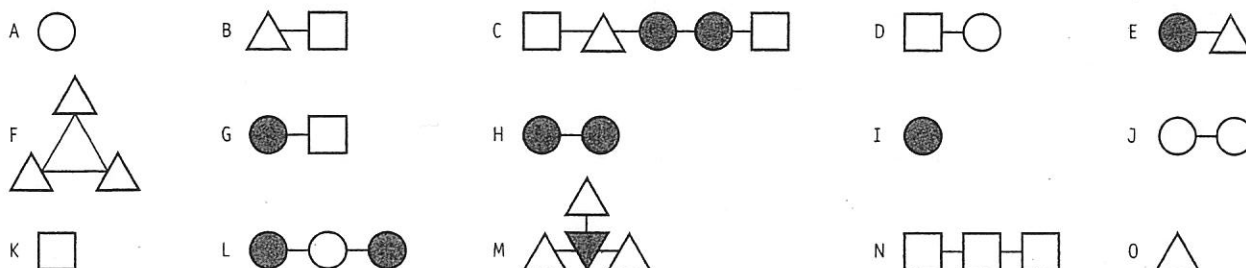


Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. _____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. _____

4



1 The following words relate to this topic about matter but are incomplete. Identify the missing letters and then use them and their matching numbers to complete the sentence below and discover the problem King Hieron had.

__ a __ tic __ es
17 6 15

m t a s
4 15

—ater
12

$$\frac{1}{4}e - \frac{1}{2}e + \frac{1}{18}t$$

13 7 13 1 16 4 19 11 15 13 8 2 7 1 16 8 3 8 4 10 15 15 1 16 4 19 11 15 13 11 6 6 4 17 15 10 5 4

[illegible]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

5

ATOMIC AND MASS NUMBERS

Write in the charge on a

- (i) proton _____
- (ii) neutron _____
- (iii) electron _____

Do you know the charge law?

Like charges _____

Unlike charges _____

Write down the name of the relationship (*attraction, repulsion*) between the following particles

proton and proton _____

proton and electron _____

proton and neutron _____

electron and electron _____

electron and neutron _____

The Atomic Number (Z) is the number of _____ in the _____

The Mass Number (A) is the number of _____ in the _____

Complete the following table

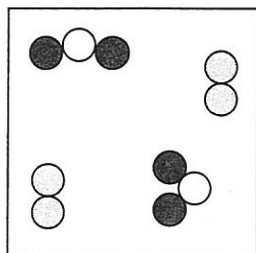
SUBSTANCE	NAME	MASS NUMBER (A)	ATOMIC NUMBER (Z) (No. OF PROTONS)	No. OF NEUTRONS (N)
$^{12}_6\text{C}$				
N		14		7
O			8	8
P		31	15	
$^{35}_{17}\text{Cl}$				
Ca		40		20
Fe				
Ag		108	47	



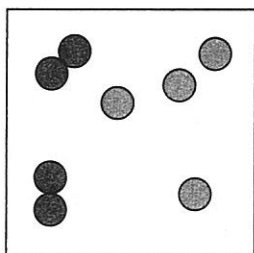
Name:

Skills: interpreting, understanding

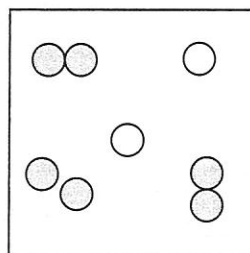
Examine the four boxes below. Each circle represents an atom and the shading pattern represents different types of atoms.



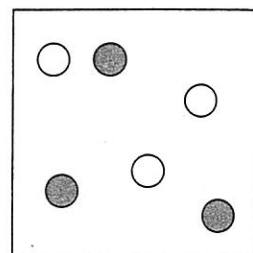
A



B



C



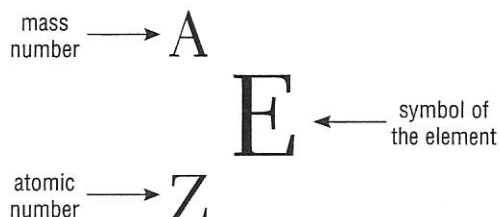
□

Questions

- Which of the boxes contains the following (note: it is possible that none of the boxes contain the requested item—if this is the case then write 'none'. It is also possible that more than one box may contain the requested item):
 - individual atoms of elements only? _____
 - chemical compounds only? _____
 - chemical elements only? _____
 - a mixture of compounds and elements? _____
- When two atoms join together we can use the term 'diatomic molecule' to describe the molecule formed (from di = 2). When a chemical element exists as single atoms, the term 'monatomic' (from mon = 1) is used. Using the samples in boxes A to D above, draw examples of the following:
 - two different types of diatomic elemental molecules
 - two different monatomic elements
 - a chemical compound

Name: **There has to be an EAZier way to represent atoms!**

There is an easy way to represent atoms so that we can describe the number of protons, neutrons and electrons in any nucleus. This method is summarised by using the letters E to represent the element symbol, A to represent the mass number (number of protons and neutrons in the nucleus) and the atomic number (number of protons).



Use this method to complete the last column in the following table. A couple of examples have been done for you.

Atom	Atomic number (number of protons)	Number of neutrons	Mass number	${}^A_Z\text{E}$
Hydrogen	1	0	1	
Hydrogen-2 (deuterium)	1	1	2	${}^2_1\text{H}$
Carbon	6	6	12	
Oxygen	8	8	16	
Lead	82	126	208	${}^{208}_{82}\text{Pb}$
Uranium	92	146	238	

Extension

Given that for neutral atoms the number of electrons in the electron cloud is equal to the number of protons in the nucleus, add an extra column to the end of the table above to include the number of electrons you would find in a neutral atom in the examples above.

Atomic Structure Worksheet

Fill in the blanks for the elements in this chart. For the purposes of this chart, round all atomic masses to the nearest whole number.

Element	Number of Protons	Number of Neutrons	Number of Electrons	Atomic Mass	Atomic Number
lithium					
carbon					
chlorine					
silver					
lead					
calcium					
tantalum					
radium					
samarium					
uranium					
americium					
lawrencium					